[00:09] 发言人 1: I think about percentage of your of the grade of this, and then we will have another part on the. Milestone，okay，so what this milestone is，is basically，um，在 the first time we announce the project，what we learn from this project is most of the group start their project really late。Like one week before you or even three days before you and try to rush it.

[00:42] 发言人 1: So we feel that may not be good. So we encourage you to start early, plan early, and start to see if your original plan goes well or you’ll notice anything that you revise. And that’s why we put a small.

[00:59] 发言人 1: 啊，last so，so that’s I think，um，if you look at schedule。That should be roughly one month or every three weeks before the project activity. Okay, so for that you need to show us that you have started.

[01:21] 发言人 1: That’s the second part, and the last three is the project itself, the final report that we expect us. Okay, so this is what you will be looking at in terms of the project formality, but like I said, the details what you need. Want you to or we suggest you to do.

[01:45] 发言人 1: So the whole detail will be released next year. Okay，so if no question，we will switch here to the to the course materials this week，okay。

[02:31] 发言人 1: So for the first two weeks, we have been talking about search problems, and the trend will continue for this week. Now we switch here to a special kind of search problem called constraints satisfaction problem. So in the first part of today’s lecture, we’re going to use two main things: one is the concept.

[02:51] 发言人 1: So what’s the definition of what is actually a constraints satisfaction problem? And then we will introduce a, just like in the previous class, we start. Responding for the search, where they do not put any problem information into solving the problem itself.

[03:07] 发言人 1: And in this case for constraints like structure, but algorithm is called backtracking. Then in the second part of the lecture we will show you how this all informed search can be made in form. Okay, so what kind of problem information that we can use for.

[03:25] 发言人 1: To make this backtracking out into box, okay? So this is the last of it. So if we take step back and think about the search that we have been talking about so far, okay?

[03:39] 发言人 1: So actually we have a view very important assumptions, okay? So first of all, you just do remember we mentioned this at every class is. Never solve the problem or search the real world problem as this.

[03:53] 发言人 1: We always have the model of the world, and we assume there’s only one single agent, which is deterministic. Right? Whatever you want to do, as a career, you will have. Okay，and every space space is discrete，okay，so we can actually, so remember the time complexity，space complexity，all these are also relevant to the process，well，at the end of the first class，we also show you the small pattern problems，so we have service space you have。

[04:22] 发言人 1: So now, if you thinking about the solution and the outcome of the search problem that we have been talking about so far, really what we want is the path or the planning. Of the problem, okay? So remember, we have been telling you the solution to a search problem is the sequence of actions that you from a starting space to the goal space.

[04:49] 发言人 1: Okay, and this is basically what we have been carving so far. And given this in mind, so we talk about it. 因为。Any node is constant, and we basically have a set method like the desk research, breakfast research, even give research.

[05:09] 发言人 1: When the cost going for one node to the other has variable cost, then we talk about the uniform cost search, we talk about a start pretty service like us, and how you can use probably information be formulated as a heuristic to speed up the problem. Right, so now in addition to this type of search problem where you hear about how you come from the beginning to the end, there’s another type of search problem which you will later on see more. There’s a more general formulation of this type of search problem or optimization problem.

[05:47] 发言人 1: But in the end, what they really care is what we know as the identification, which means is the goal itself is something you care. Okay, so if you. Think about this, and original probably previous search probably talk to you, you actually know the goal, you can do a goal test whether the goal actually whether what you go but the destination wherever you want to be is actually equal to the goal itself.

[06:18] 发言人 1: So the path is something that there’s another type of search problem where you do not know, for example, you want to design something that satisfy some design criteria, for example, you want to design a room that can fit150 students okay? And how you should design the room you do not know but you have some constraints. You have some requirements that you won’t fulfill.

[06:44] 发言人 1: And this type of problem we call identification problem. Ok, so what really matter here is the goal itself. how this this whole look like you only know what you want but you do not know what happened design. Give you what you want, okay?

[07:02] 发言人 1: Like you designing a bridge, you know how much maximum money you want to spend, you know how long the bridge needs to cover, you have some design criteria says how many cars and what things we pass through without breaking your your your bridge. For example, then how exactly is this bridge look like? Okay，so this is another type we call identification problem，and constraint satisfaction is actually what type of this identification problem，and a very special type of identification problem。

[07:37] 发言人 1: Okay，so now we know where we coming from，right? From the beginning now，then let’s dive into the after definition of the constraints that. Okay，so basically，if you’re thinking about a standard search problem，you do not really understand the internal dynamics of the world，right?

[07:59] 发言人 1: So you take. Action, you have the success function, which tells you if your action was the cause of how your world is going to transition, but internally you do not know if that happens. Okay, but if you’re looking at the satisfaction problem, it is actually way better.

[08:19] 发言人 1: Well defined. You can consider from this perspective. Okay, so it is first of all, it is a subset. Generous and problem and identification problem okay and it is define very. Could you mean by the variable that you try to identify for x.

[08:40] 发言人 1: Okay，it is usually a vector，so there’s many many subjects you want to identify，and you know there’s a domain for each one of these。Okay, it the way basically means what are the evil or what are the feasible values for each variable that you can take. Okay，so if you thinking about this，so we will look like we have x as a vector。

[09:12] 发言人 1: Okay，and this is what we call variables。That will basically display the design. So if you thinking about this as a scheduling.

[09:25] 发言人 1: OK，and this x could be which professor take more class。OK，for example，X one is math class，you can assign this math class to some professor from a domain。Let’s say for math class X one, you have professor A, B, and C.

[09:46] 发言人 1: We can teach. Okay. Now for x two which may be english class you have another set of professor baby b and c and. You can teach English class, okay? So on and so forth. So a constraint satisfaction problem is what we want to find out the active variables be labeled by x, and each of this decision that you want to figure out has a set of finite choices available, and that is called a domain. Okay，then the goal task is a set of constraint，a specified allowable combination values for such self set variables。

[10:30] 发言人 1: So given this example，what it means is that you have some constraints，but this class teaching assignment problem，you can figure out the constraint first of all，all class needs to have the feature。OK，second，a single teacher cannot simultaneously teach two classes that happening at the same time。As physically possible，third is you only have qualified teacher for every class，you cannot。

[11:03] 发言人 1: Okay，so if you satisfy all these. The NBA stated gold test were industrialized store problems checked. ok.

[11:13] 发言人 1: So far so good right any free comments or free questions. OK，so if we thinking about this，so if you compare。This constrains satisfaction to a general search problem.

[11:29] 发言人 1: We have better defined. We have actually more information about the problem. So actually the algorithm itself is to take the taylor to solve is more efficient.

[11:40] 发言人 1: Ok, so just through. Help you understand this is a straight satisfaction for this class. We will using the following running sample for the test book.

[11:52] 发言人 1: The AI model approach book, which is called color the map coloring chart. So this problem goes like this. First of all, what you see here is the map of Australia.

[12:04] 发言人 1: Okay, and it has been divided into one, two, three, four, five, six, seven, seven different states, just like the problems of our country. It has seven different states. Okay, and the problem asks us to do reform.

[12:19] 发言人 1: Okay, so if I can color each state in one of the three colors, red, blue or green, okay, can you give me a particular coloring plan such that there's no two neighboring states share the same color? For example, if you look at here, the western Australia state and the northern territory state are the neighboring two states. It says they cannot be covered the same color.

[12:53] 发言人 1: Same goes with western Australia to South Australia. They are also two neighboring states. It says they cannot be covered this. If you look at Western Australia to Tanzania or Tanzania to Tasmania, I mean to New South Wales, they are not every state, so we can be color the same.

[13:12] 发言人 1: Okay, so the problem is you want to paint the color for every state in Australia such that no two neighboring space share the same color. Okay, so how can you actually solve? So now, if you thinking about a constraint satisfaction problem CSP has at least two main things.

[13:34] 发言人 1: One is what are the variables that we have to do. Okay, so in this case, the assignment identification problem, the one that you try to identify is the very。Okay, so in here, the identification is basically the color of each space。

[13:53] 发言人 1: Okay, so if you count, there are seven different states。Here that means we have seven different variables. Each one corresponds to each state. So we have, we are now using abbreviations. So W stands for Western Australia, NT stands for Northern Territory, Q stands for Queensland, NSW stands for North, New South Wales, VB stands for Victoria, SA stands for South Australia, and T stands for Tasmania.

[14:24] 发言人 1: okay these are the seven decision variables that you want to figure out what color you should choose okay and that is your very and in this case the domain is free color red green and blue okay something be careful about this in this particular problem the domain for every variable is the same. So every state can choose one of the three. But in many real world problems, the domain of different variable can be different.

[14:56] 发言人 1: And in reality, that's most likely the case. Every variable will have different choices. Okay, like which professor can teach what course is going to change depending on the course of this.

[15:08] 发言人 1: Okay, so now the constraint we want to check on the gold testing, and we need to test is there's no two adjacent regions or two no two neighboring states have the same color, okay, or they must have different color。Okay, so now if you write down these constraints mathematically, there's two ways how we can write it. The first way is we call implicit.

[15:39] 发言人 1: Okay, so basically what this constraint basically means is that the note that the western Australia cannot be equal to the color of western Australia cannot be equal to northern territory, and the northern territory cannot be equal to

Queensland，and Queensland cannot be equal to New South Wales。And the news comes well. And not be equal to Victoria, so on and so forth.

[16:06] 发言人 1: Okay, so why this is called implicit? Is you did not provide a specific values to each one of these constraint value, but to this variables all you provide is some relationship. Okay，so as long as they are not equal，they are not telling you what value can be or value cannot be。

[16:30] 发言人 1: Okay，and this is what we call an implicit。And then there’s another way of defining this constraint is called explicit, where in this case you explicitly write out the feasible values for each one of these. Okay, so now if you’re thinking about the color combination for Western Australia and the Northern Territory, what they can do is if.

[16:56] 发言人 1: Your western could have a color of red. Then your northern territory can either be green or blue. It can no longer be red anymore.

[17:08] 发言人 1: Okay, now if you think, if you’re not, if you’re Western Australia, now take a color of green. Then the northern territory can either take a. All blues, right? You cannot pick any color. So now, if your Western Australia take the color blue, then your Northern Territory can even take red or green. Okay，and now we move on to the next pair。

[17:38] 发言人 1: If the next pair you’re looking at is northern territory and the Queensland，if your Queensland，if your northern territory take the color of red。Then your prince like can either take green or blue. If the northern territory is a follow green they can even be red often so on and so.

[18:01] 发言人 1: Right，so now why we call this explicit，I hope this already obvious to you。Now instead of specifying the constraint as a relation，you actually gave up all the feasible values，the actual values。Okay，but what problem you will immediately notice is that implicit constraint definition is made way more compact。

[18:26] 发言人 1: Right，because you can income，you basically anything，everything you have seen so far。Can be represent just by the single equation of W A not equal to F B. But if you write down explicitly, you will have a really really long list.

[18:45] 发言人 1: Okay，so if you are working with the problems many many variables，they explicit way of writing it，they will be very long。Okay，so either way is fine as long as as long as you have specified everything。Okay，now have this specific after we define this CSP then。You will think about what’s the solution to this problem.

[19:08] 发言人 1: I think the solution to this problem after you knowing the definition should be clear, which is basically a particular set of values that you assign to every single one of these variable such that all the strings that okay so in this case if you assign the color like. 这是啊。严格实施推进的申请。

[19:31] 发言人 1: Okay，for example，if Western Australia is red，London airport is green，Queensland is red，New South Wales is green，and Victoria is red，South Australia is blue，Tasmania is green，and South Wales is blue。No two neighboring states share the same color。And each one of these has been assigned.

[19:53] 发言人 1: okay，so. If you thinking about this problem particularly for any CSP problem, do you think the solution going to be unique? Can we have multiple solution to this problem? Maybe can we have a different assignment such that every state have a color and no two neighboring space share the same color. What do you think?

[20:33] 发言人 1: Okay，um，I think the problem is answer is yes，because in here all we are asking is as long as no neighboring station，so it will be fine，so you can change westwestern Australia to different color Starbucks。And you can have a new step sequence of colors, besides my own. Okay, so now if you think about this color, this this probably particular. how many different solutions do you think is there?

[21:24] 发言人 1: Any volunteers want to give a try? A 厅，A 厅，Why yesterday? Chinese 比 OK，Chinese 比 OK。就是我说中文可以吗？中文我说我说中文可以吗？可以可以啊，就是那个。右下角那个就是三种可能性嘛，对，然后呃，趋势。上面的那个大的板块里面随便挑一个，就是它有三种可能性可以选，然后它的相邻就两种可能性可以选，所以三乘二乘三，三乘二乘三，所以等于 18，等于 18，ok。

[22:20] 发言人 1: I need to get back to your logic, but the logic I have is the Western Australia can choose one of the three. Start with because that’s if we start with western Australia and this is the one they can freely choose one of the three. But each choice that you have, the other people in the remaining state needs to choose every other options.

[22:42] 发言人 1: Okay, so in the end, my answer equal to yours. Was it as media if we choose three of them, so in that is actually eighteen, right? But I think the logic may be right or even more clean to minus one. Then think about this together. But the answer is this eighteen.

[23:00] 发言人 1: Okay，so in general，this is true for any CSP problems，you can actually have multiple solutions，and the reason，right，for this multiple solution to occur，is you do not have a problem across all these solutions。All you are asking is whether your first friend specify or not. Okay, so usually there is not a, there is more than one way to specify.

[23:27] 发言人 1: Usually, Okay，so later on you will see in the next lecture，also the next after next，you will see a more general formulation on optimization，where in addition to this constraint，you also have some preferences。Okay，also you can imagine this preferences to solve constraints。Okay，so let me continue，so now you understand。

[23:50] 发言人 1: The definition of the C S P now you see an actual example of C S P yeah we going into different kinds. So what kind of CSPs are there? So the most simple CSP we can have is called a binary system, where every constraint only involve two variables.

[24:14] 发言人 1: Okay, so now if you try to understand what this means, we can think about this as a constraint. Okay，what in this graph every node is a variable that you try to figure out。So for this case，it is the one space from the Australia that you need to figure out the problem。

[24:33] 发言人 1: Okay，and if there’s a constraint that relates any two nodes or any two variable，you put an arc connecting them。Okay, that means if in this problem if there are two neighboring states you have a R connecting. And in the second part of today’s lecture you will see that if you formulate this.

[25:00] 发言人 1: Problem as a graph structure, there’s some useful structure information from this representation that you can leverage. The song is from much faster. For example, if you put this into a graph, you will see that because media is disconnected from everything else.

[25:23] 发言人 1: That means, whatever very good decision that you need to consider. You can basically leave out this paper of any of these constraints because it is not connected with anyone else, so it can be which is shown over here. So if you see here, basically what you capture is first of all you will have one variable for every state, so for this problem you should have an overall setup.

[25:50] 发言人 1: Okay，so now you see there’s an arc connecting some of the，that basically with black geometric neighboring two states，you also。For example, West Nile Street is connected is an eight base thing to not the territories who here is on. Okay，so after you。

[26:10] 发言人 1: Put them into a graph like this, you will immediately see Tasmania is the one by hanging around not connected to anything. Ok, so this is a visual representation of this. So if you have.

[26:26] 发言人 1: Learn graph structure, maybe from entry or data structure. You can actually put this into a Jesus matrix. Okay, so what we will show up is, let’s say now we start with WA to Tasmania where there are several of these, and here also.

[26:47] 发言人 1: Western Australia Tasmania，and this matrix tells you the connectivity of the everythings。So now you can see the matrix，what you will notice is that because media is not connected to everybody。So the whole column and the whole row connected to Tasmania is actually zero.

[27:06] 发言人 1: OK，so that’s how you can tell this is big data，because in reality, if you imagine，if you have a hundred different variables。Okay，and maybe every constraint involve more than two，then this graph by joining out can be really really complicated to visualize，but if you put this into a matrix. Doesn’t matter how many row and column, that’s just an by an, and equal to number variables.

[27:35] 发言人 1: If you have a million, that’s just a million by a million. All you have to do is some all the rows and columns, you will know whether there’s any connect to it or not. Okay，so it’s actually very easy for you to do that.

[27:50] 发言人 1: Okay，so now let me give you one more really example on how this, how this CSP can be used. Help you solve some questions, some games for example. Okay, like I explained in the first lecture, AI particularly the AI before deep learning era, people are fascinated about solving games.

[28:14] 发言人 1: Okay, this end queen games is one of the early games that people try to solve using CSP. Okay，so what this game trying to say is n equal to the size of your

chestboard。Okay，so if you see here，you have four row by four column，so this game the n equal to.

[28:36] 发言人 1: Cool，okay，so when you have a whole life for chess board，anything game helps you place anything in this case four things exactly into this chess board，so that。There’s no two queen at the same roles. no two queen.

[28:58] 发言人 1: in the same column. No two queen in the diagonals as well. Okay, so if you play chess before, you know the rule of the screen is it can kill any queen in the same row or in the same column or in the diagonal.

[29:19] 发言人 1: Okay，so this game ask you to place exactly n，where n equals twice the chess board，such that no two queen can kill each other。okay，so now you thinking about this。HOW CAN I FORMULATE THIS INTO A CSP PROBLEM? SO, FROM THE PREVIOUS EXAMPLE, YOU KNOW, A CSP PROBLEM NEEDS TO DEFINE THE VARIABLES, NEEDS TO DEFINE THE DOMAINS, NEEDS TO DEFINE ALL THE CONSTRAINTS.

[29:51] 发言人 1: Then you have a solution, right? So for this case, you need to define the very, what’s the variables, what’s the domains for each one of these variables. And while the customers, okay? So here we can define this at least using two different ways.

[30:07] 发言人 1: And the first way says: if now, let me do this, if I now use two indices, okay? Where the first indices. I tells you which role you are currently in.

[30:23] 发言人 1: And the J tells you what column you are currently in. Okay, so now what you want have is. The variable is basically Xi, where I equal to number of rows here equal to4, J also equal to the number of columns.

[30:43] 发言人 1: Now here is1 to4, okay? And what you want to have is you want to solve every element in this matrix, okay? So basically XiJ in this case is a4 by4 matrix.

[30:58] 发言人 1: You want to sum this matrix to be exactly equal to n, which captures you exactly put any place into this transport. Okay，now the domain is basically tells you in this particular location，so this matrix is basically. Imagine this to be the matrix I energy every domain basically tells you every entry wants the number in the cake。

[31:29] 发言人 1: so you can either have a queen. Meaning zero, um, no, zero meaning no queen. Or a one means you place a queen here.

[31:43] 发言人 1: Here’s the queen at this location. So x zero zero means, because I we are using the python in that patient x zero zero. If this equal to zero, that means there’s no point.

[31:59] 发言人 1: If x zero. Because zero equal to one, then you have a point there. So you know in this case this equal to zero and x zero one equal one because there’s a point right here.

[32:11] 发言人 1: Okay, that’s why I previously said if you stop everything together, you should exactly what we have. Because you should exactly have four ways. ok.

[32:23] 发言人 1: So now we understand how we set the variables and the means for each one of the variable. Now let’ s take a look at the constraints, what they actually is. Okay, now for you, for us to express this constraint, we introduce another.

[32:42] 发言人 1: Industries called cake. Okay, let me explain what it means. Now, if you look at this combination, X I J and X I K.

[32:55] 发言人 1: Okay, what he says is these two location are in the. Same role, that’ s why their first index is also always high. Okay, and J and K are two different indices, that means they are not exactly the same location.

[33:14] 发言人 1: So these two entries are at the same row. 啊， two different calls。 Okay， and you can either take the value zero and the zero。Or zero one or one zero.

[33:31] 发言人 1: So what it means is for two cells in the same row. Okay， two different cells in the same row， you can have me the no queen。What exactly one thing you cannot have one and one so equal to two.

[33:53] 发言人 1: So for two different cells in the same floor, you can only have one thing on. Okay， later on you will see that if you put this， you should not solve the problem， but just as far as the constraint is satisfied， you can either have no way or two。I mean there’ s no way exactly one.

[34:17] 发言人 1: Okay, now if you move on to the second row over here. We share the same index of Holland J. That means for the same column, two different cell can either have no queen.

[34:35] 发言人 1: Or exactly one queen. Okay, so if you think about the screen, basically constrained there’ s no two queen in the row direction, then this second is basically constrained there’ s no two queen in the column direction. And now, if you see the last.

[34:57] 发言人 1: you x and y did not explain more and quality exactly be offset by a positive. Hey, in this, that basically tells you there should be no queen. In the sky. And the last one, they all be. A start like this, just basically this diagonal right here.

[35:29] 发言人 1: Okay， so if you take this definition again， he says， I’ m going to model this problem exactly equal to the number of locations， the number of cells in this chessboard， where each cell can contain eight ways of not having the pool。So that’ s the way a variable is each one in this chest. Okay， then the constraints says no two queen in the same row， no two queen in the same column， no two queen in the diagonals。And if you sum up all the things, the entries numbered ones in the chessboard, that should exactly equal the answer.

[36:19] 发言人 1: Okay， so this is one way of how you can express this equation. Problem, okay? So now, if you think about, if you look at this formulation, it’ s a little bit verbose, right? It’ s quite elaborate, okay?

[36:34] 发言人 1: So now, the second formulation is how can we utilize this more problem information in the compact way, okay? So it says, if you remember this problem, it says there’ s no two queen in the same row or the same column. That’ s basically, if you think about it, that basically means you should exactly have one bringing every role and every call.

[37:00] 发言人 1: Otherwise，there’s no way you can have exactly any queen in the chessboard without completing the same row or in the same column。Right, if you remember, if you try to remove anything from here, if you have any goal or any column without a queen. Then you must have to bring it either the same row or in the same column, so you cannot satisfy this problem, right? So now, if you modify this problem, changing the definition a little bit, says I will have exactly one queen in every row and every column.

[37:37] 发言人 1: In this case, we said we will have exactly one queen in the first row, exactly one queen in the second row. Exactly one point the third, exactly exactly one point the fourth as well. Then the problem becomes: which one? Which? Jay, this is which column in the first row? Is your first way and which one out of the second row which Jake index for the second row between it is your activity you place the question okay so your.

[38:17] 发言人 1: Variable now changes from the x and y location to one earthquake. in domain. It’s basically domain where j can pick so for any problem you can pick from one to n.

[38:37] 发言人 1: So for Q one, it basically says I will place the first queen in the first row. The decision is basically in which column I should place. And the domain one to end will tell me exactly which column I want to put.

[38:57] 发言人 1: ok. Any question on this one? Now if you formulate this, then the constraint becomes way simpler. If remember Q one always stay in the first row, Q two always stays in the second row, then the available combination is if you place the first ring at the same as the first column. Then your second queen cannot be at if your Q one is right here.

[39:31] 发言人 1: Then your Q twenty two cannot be at the first column, cannot be at the second column because that will be in the diagonal. The only possible one is the third or the fourth. Okay，these two you can choose。So now，if you move your Q two to here。

[39:56] 发言人 1: Then the possible location, if you move your Q one to the second. 好，点的 legal location for your Q2 is simply here。because you cannot have the same.

[40:14] 发言人 1: column and diving. Okay，so that means if this is two，then only three locations for. Again, if your Q one is.

[40:29] 发言人 1: Here, then the only people location is one. So this is three, then the only people location is one. Likewise, if you put your Q one here, then the people location is one.

[40:46] 发言人 1: or two. So on and so forth. So now you can move on if your Q two and Q three and Q three and Q four, etc.

[40:59] 发言人 1: Okay，any problem with this? So you can see what we want to show you through this example is not only the solution。To your CSP problem is not unique.

[41:13] 发言人 1: Also, the definition how you define the problem is also not unique. And oftentimes you can see different definition, different how you model the

problem, how you define the problem has a huge impact in the following solving. Okay，and this will be true for the project you do。

[41:33] 发言人 1: So most of the project that you want，that we like you to try to solve, will have a different base of model，and your modeling will have an impact in a subsequent solve of that。Okay，so now Nexus，we we we kind of implements more of the same into the H B M L for you to try it out。Okay，so what you have，what you basically want to try is you can place anything。

[42:00] 发言人 1: Any location of this, then you will show you all the things that the market brand is the one that is illegal. Okay, then you can try this yourself in the end to see if you can figure out the way of solving this. Okay, so you can try this or you can try this with an algorithm.

[42:19] 发言人 1: Okay，so this is basically the example that we provide now, especially can. Try to summarize the summarize different varieties of csv problem that you can have. Okay, so first of all, we can define this from the perspective of variable.

[42:43] 发言人 1: Okay, so now if you trying to looking at the variables and if your variable can take finite values. What it means is the domains for each one of these variable can be is finite, then basically you will have the tone. Number combination that you’ll have with the C three problem is basically erased from our and and this number variables.

[43:09] 发言人 1: okay，and the thing is the number of choices you have per day。Okay，so。In this case, even a boolean satisfaction problem, a boolean CSP. So every value can either be0 or1, can be empty hard, or more precisely empty complete. So there’s no polynomial time algorithm as well.

[43:35] 发言人 1: Okay so just a little bit context you might have heard about the empty from the news and because of this A size people have been talking about this quite often so for those who do not understand and let me give you small intuit what he actually makes because you can go to the algorithm class you will hear this very often so basically what does N versus P means is this so very very long time. There’s a mathematician, so you don’t have to remember the name, but what he have a hypothesis is follows. So there’s many problems where you can very very quickly verify if a solution is feasible or not.

[44:15] 发言人 1: Like in the end game case, I gave you a particular location of this end game. You can very quickly tell me that is okay, right? Very very quick.

[44:26] 发言人 1: That goes with the routing, the routing assignment, which machine needs to manufacture this for that. All these assignment problems, any of these problems are very very quick. If I gave you a solution, you very quickly can help me there.

[44:39] 发言人 1: So this expectation has a hypothesis that if the problem is so quick to verify, that’s also means that the problem can be very quick to solve. Because you can very quickly test whether it is working or not. That means the problem can be very quickly solved.

[44:58] 发言人 1: And this is the problem behind. So if the problem is very quickly verified, does it also be very quickly resolved? And so far, we have not have a definite answer.

[45:09] 发言人 1: If you know the answer to it. Very glad you will likely be a Nobel Prize. But basically this is basically basically means for this type problem even though it is very quickly verified but finding a solution is still very very good.

[45:25] 发言人 1: OK, and you might not have heard about this boolean satisfaction problem before but this is actually the fundamentals of computer science. Almost every basic fundamentals of the problem formulated as a boolean set. Okay, so if you nowadays now thanks for AI must have heard about news like different ships all these companies are practicing new ships beyond Libya Huawei making their own ships.

[45:57] 发言人 1: But when you thinking about ship design. Okay, so all these EDA, all these chip design problem in the end is solving it. So it is the CSP.

[46:12] 发言人 1: Okay, so if you know how to solve this well, you can join the company who manufacturer chips. okay. So now you might have some CSP problems where you have infinite.

[46:26] 发言人 1: Okay，the number of choices that you have is basically in by that you cannot count it，like the most of the schedule account。Okay，basically the variable is when this job should start。Okay，it can be any time in the future，there’s no really about to。

[46:44] 发言人 1: Okay。So now you may also have continuous variable, so the variable a continuous values like like the schedule when this is actually start okay? So now you need to find the CSP from the.

[47:00] 发言人 1: From the perspective of constraint, there’s many ways how you can define that as okay. So one of the very simple CSP is every constraint only involve a single way, and we call this binary. Concerns so any example is you specifically tell what option is available for this variable one so the whole computation only involves one simple there.

[47:32] 发言人 1: And binary constraint is basically the constraints that involve two way. ok，for example，it says the color of southern australia cannot be equal to the color of western australia。That is a simple constraint involving two variables, and therefore unary and binary constraints.

[47:53] 发言人 1: That’s actually problem we know very efficient algorithm to solve them, but unfortunately. If you go to more complex CSP, like the constraint involving more than two, or the relationship of this constraint is more than linear, then we do not know a good algorithm that have guaranteed to solve this problem very efficiently. Okay, so actually, if you are studying computer graphics, many of these computer graphics problems is CSP that involve more than three variables, like in the geometry perspective.

[48:31] 发言人 1: Okay, so on top of these different type of constraints, we have another special type of constraint, which is so sometimes known as soft constraints.

Or practice, okay? So if you're thinking about, if you take a look at these constraints that we have been talking about so far, they are what we call hard constraints.

[48:51] 发言人 1: This says you cannot do this, or only this is feasible, you cannot do that, okay? And there's another type. CONSTRAINT NOT BE SO STRICT.

[49:03] 发言人 1: there are more or less like you know if you do this is bad if you choose this it is even better ok so you have some ordinal relationship which we know as elastic graphical basically says for example if you color the space with red it is better if you color this. For example, we have some practice, ok? So later on, you will see that this preferences in the end can be formulated as an objective.

[49:31] 发言人 1: Okay, and that will leads us if you have for problem have both soft constraint and hard constraint, then you basically encounter a general optimization problem, okay, which we will cover in the following lectures, okay。So now, like I said, I've been giving you some example of the robot CSP problem, right? It is actually there in almost all of these commercial research。OK, 那 schedule 你的 problem, type big。

[50:00] 发言人 1: In the end, if you're thinking about the university, they have lots of CSP to solve scheduling classes, scheduling final exams, except these are all scheduling problems. Okay, or assignment problems, professor or teacher, or courses, and hardware constraint, hardware configurations. Okay, and this is now one of the very important problem for large language model service company, like for example the seed, the finance and all these online providers。

[50:34] 发言人 1: Now you have many requests sending at the same time, so which hardware, which server actually should serve what threads? So in the end, this is a constraints I practice problem, because internally they always constraints us every query on the normal networking。conditions should give you a feedback within two sentences.

[50:55] 发言人 1: Okay, now the rest of assignment is this very hard to reach GPU for increase。Okay, and transportation scheduling, factory, and certainly how this should design, okay, so all these are actually. Okay, so now, um, after all this, let's take a quick ten minutes break, let's zoom back around ten thirty five, and we will start talking about how we resolve this year. Okay.

[51:45] 发言人 1: 这个计划书这种颜色的有时候有没有问题啊这种颜色对就是这种表格不是图片对图片变成这样子没有问题没有问题后面的还在改我先装然后这个。字体，然后格式对齐什么的，我在后面再调一下，你弄好就。嗯，我先交校稿，嗯，校稿不至于错，格式，还没法格式。校稿弄好看点，就按照论文来吧，这样好看一点。我到时候不再帮你调一下。

[53:00] 发言人 1: 嗯嗯，所以你说的这个交叉运算，这个部分模型是，就是有像这样，就是你需要的，对吧？我就感觉就是不确定的，对，导致这个他的二零出来，他他如果是输出区别，就是 IGP 这三个位置，我们自己能有的表是算出来的。

[54:20] 发言人 1: 因那这当到这 K 个话这那这个的时就就指的是这两。是的，因为这个事他有，那具体怎么做后面也是可以。如果说他也不能，你至少几点要不就王夫这个地方，那是不是他没有，他有可能就都没有什么情况。然后至于这为什么配的这个其实就比较有意思，因为这个是斜线。这这个细节就是 IG 比如说 IG 是 IG 是这一个然后这个是两个是两个你反正其实后面这个都一样那这一排意思是什么就是就是。就是这一道电只有一个这块就是这个意思然后

然后这个还是斜着走就是比如说这个还可以加 K 就是 K 二 K 二的话有更多很多都加 K 嗯这个。

[57:05] 发言人 1: 啊，对。

[57:33] 发言人 1: 讨论。

[58:02] 发言人 1: 共。

[58:21] 发言人 1: 还有五分钟。

[58:36] 发言人 1: 认证。

[58:54] 发言人 1: 你告诉别人。

[59:08] 发言人 1: 开启。

[59:44] 发言人 1: 看不清。

[62:29] 发言人 1: So we said please demonstrate as we zoom back to talk about how we can solve a CSP problem. Okay, so basically we have been talking about a few algorithms before, right? We talk about.

[62:44] 发言人 1: That’s first search, that’s first search, you create deeply uniform calls and et cetera. So the first ideas can be use any of these to solve this problem. Okay, so now when you are trying to use these existing tree-based search method, then you need the first three.

[63:00] 发言人 1: You know what is actually the Thursday, we actually the three days right? So now, if you are thinking about CSP, basically the initial group that you will have is an MVP. where none of these variables has been solved, so if you’re thinking about the searchy and the root note.

[63:27] 发言人 1: The service is basically an empty set where none of the variable has been used. OK, then the successor function is basically assign a variable to an outside, basically assign a value to any outside variable. OK，所以你想想 solution to the C F G，which order you visit all these very well，really does that。

[63:51] 发言人 1: Okay，and the last thing when you do this go test is basically is all these variable has been assigned，so seven variable in the Australia game。At least all seven have been assigned, you check if all the constraint has been set. If that’s the case, then you’re done.

[64:08] 发言人 1: Okay, so now what we want to do is start with the very basic search that we have been talking about and see how they will perform. Okay, so now if you’re thinking about the grasp search. Okay，so if you thinking about this cartoon.

[64:28] 发言人 1: Of press for search，so what it does is that it will visit all the nodes before a particular depth，then increase the depth like this。Right, so now if you are thinking about the CSP asset empathy. 这三门还是如此。and.

[64:52] 发言人 1: THIS IS OUR. That’s when that’s M, all of these solutions will only occur. At the maximum depth, so if we put this into CSP’s language, this depth is basically number variable and and all the solution will exactly happen at the largest depth.

[65:17] 发言人 1: That’s where all these society has been, not the very wise being assigned about. OK，So now if you apply a grasp for search algorithm to solve，you will be extremely inefficient。Because it technically visit everything before you can see a solution.

[65:37] 发言人 1: Right，all the solution can only occur at the maximum depth, that’s where all the variable has been assigned。The store is empty，and gradually assign a variable，so here you have one value，here you have another value。etc.

[65:53] 发言人 1: In the end, this is where you have all the end values. So at first glance. Without knowing any details, we can immediately know or immediately predict grasper search is going to be really bad, at least really, really inefficient.

[66:10] 发言人 1: You will basically visit everything before he has any chance of seeing a solution because he actually visit every depths or the possibilities before the next event. Okay, so the conclusion is very easy. I hope you you can immediately notice that now this is not going to be the this problem.

[66:31] 发言人 1: Okay, so now you are thinking about the that person. IS THAT RIGHT? 你还要注意，visit some part of the tree all the way to the maximum depth。Before you move on to another part, like this. Right, so if we know.

[67:03] 发言人 1: Your solution to your CSP can only occur at the maximum depth. The depth for search has a hole, at least much better than the best for search. Okay, and at the same time you know that this including both depth first and the best first, they do not incorporate any problem information.

[67:24] 发言人 1: So even though the depth first search is better, but still very inefficient. For this particular CSP problem, okay? So that’s why in the following we are going to show you a much better algorithm, hopefully a much better algorithm that’s the plane that’s for search.

[67:42] 发言人 1: Okay, so we will move on to the second part of today’s lecture on how you can solve CSP better by building on top of depth-first search. ok，because we know BMS really that。OK，so in particular，we were talking about three enhancement，which is watering。

[68:05] 发言人 1: Filtering in the problem structure, okay? So these are the problem information that we use to make it run much faster. You just... you can imagine these three to be the heuristics. About CSP specific human space that we use to make it from us.

[68:27] 发言人 1: Because you know from the previous lecture, all these search algorithm, as soon as you frame the search algorithm like search tree based problem, then the only way to make it efficient and effective is incorporating this human space. Which tells you estimate how good it because it is to your work. Okay, so now let’s start with the basic structure which is called a backtracking search, which is basically the most basic uninformed algorithm for solving a CSP.

[68:59] 发言人 1: 我。So simply put, it is a variable at the time was checking constraint here. Okay, so let me give you a little bit more details.

[69:11] 发言人 1: Okay, the backtracking algorithm start with one variable at a time. Okay, so in this case, remember we have if we are thinking about Australia case, we have Western Australia. Northern Territory, etc, Tasmania, in total seven variables.

[69:29] 发言人 1: Okay, so what this backtracking algorithm will do is you just randomly pick one area to start. Okay, because in the end, all you care is assignment to a particular solution. So which variable you start does not change the actual solution itself.

[69:47] 发言人 1: So these always work on one very well. So this backtracking algorithm will start assigning a variable. 我 starting assigning value to one of the variable then move over the x ok s was. 然后第二个事情，然后你就是 equal check constraint as you know。Okay，so later in the next slide，I’m going to show you a confidence。

[70:14] 发言人 1: Okay，so basically what it means is that as soon as you assign a value to a particular variable，you immediately check if there’s anything that’s been left for the other states to consider。Okay，and these two together on top of DFS is basically our backtracking algorithm，and empirically you can solve the entry problem that we previously show up to twenty-five points。Okay，so without these two idea one and idea two，we basically can only solve around eight points。

[70:51] 发言人 1: Okay，so let me actually give you example how we actually work again。We are using the same state。Case as an example, okay? The goal is to assign a color to each one of the state of the seven such that no two neighboring state share the same color.

[71:10] 发言人 1: Okay, so what the backtracking algorithm do is take one to start, anyone is fine. Okay, randomly take one. Let’s say now.

[71:20] 发言人 1: I start with the state of Melbourne Street, which is this way, right? Okay. So now the backtracking algorithm will have basically three options to choose from.

[71:35] 发言人 1: It can be no other state has been assigned any color, so we can choose either red, green or blue. Either one’s fine. So now just like any DFS, we just pick one.

[71:46] 发言人 1: Let’s say we go with the previous convention of always pick the one on the left. So we go with red. Okay，and that charging algorithm says after you fix W A，the first variable we move on the second。Let’s say the second variable we’re trying to fix is this northern character, this one. Okay，and previously what he says is the idea numbers number two is check constraint as you go。So what he means is that previously at every state you can choose three color check constraint as you go, is as soon as you assign a value，you check what other remaining feasible value。

[72:29] 发言人 1: Then that for you to choose, for example, after you are setting a color to this Western Australia and the second variable you’re trying to find out is northern territory, then you immediately can know if my WA is fixed at the back out of the three there’s only two remain visible which is green and blue. This is what we mean by check constraint as you go. You do not let independently choose all the options.

[72:58] 发言人 1: You only allow them to choose the one that still satisfy all the constraint. Okay，so backtracking number of just that simple take a variable anyone to start assign the value then move on to the next when you assign try to assign the value for the next variable look at the remaining feasible choices other than all the choices. either.

[73:24] 发言人 1: OK，so now let’s say these two are equally fine，so I go with the DFS convention of picking the left one，so I sign that to be green。After I signed the northern territory. And the W A is bad and green.

[73:42] 发言人 1: Now the next one I’m looking at is this one. 喂，so my queensland tell me we can choose three color red blue and green。But this queen’s like is a neighbor to this northern territory. So if I check my constraint as my goal, then the only remaining color in this case is either red or blue. Okay，and so on and so forth。

[74:11] 发言人 1: Okay，so if I go with convention of going this，then the next one maybe I can check is the southern Australia or whatever order you pick，then I look at what’s the remaining color that’s possible。Okay，and if I choose the southern Australia as my next，then you can see the only option is to take a group because it is the neighbor to Western Australia，neighbor to Northern Territory and neighbor to this Queensland。The northern territory already took green western australia and the things not already took red so the only option that is blue.

[74:49] 发言人 1: 我买四号。Okay, so this is basically what we mean by check constraint as we go. OK，any questions so far? That’s the basic that for search you see how expand is basically EMS。Okay，and the backtracking basically tells us, instead of letting you choose anyone，any option from the from the domain，we only choose the one that’s satisfied the constraint，that’s it。

[75:29] 发言人 1: Okay，so if you try to look at the suitable of this right tiny elder dog, you will have a better understanding，but in today’s explore，so in the cost use the time，so we are not going over this super cold line by line，but we have implement this。In the tutorial to the number four so you can actually go there and along this line by line to see how you actually solve ok so. What we want to show you the backtracking algorithm, what we want you to actually understand is backtracking is basically a death verse work plus the.

[76:13] 发言人 1: The checking the constraint as you go I D. OK, only stop. So, DFS will stop as you are reaching the final depth.

[76:24] 发言人 1: So, you are reaching the DFS, the original DFS. Will stop after you color all seven space. So basically you are doing this eight level here.

[76:39] 发言人 1: Right, you’ll stop. But if you consider backtracking, you will stop as soon as there’s no remaining feasible solution for you to check. Okay, so now let me, if you use this as example, you will see what I understand.

[76:53] 发言人 1: So now let’s say instead of going left, I go right. So I factor all these, then I go into right. Okay，so now in here you can see the Western Australia has been colorless red，Northern Territory has been green，Queensland has been red。

[77:11] 发言人 1: Now if I want to color this South Australia，this essay。Just know remaining color remains feasible anymore because it’s neighbor to all the three and each one of them has making a colorful like. So if you try to go to a further desk, you results in no feasible solution.

[77:35] 发言人 1: So you stop, you don’t have to go further at least for this branch. Then combine to this one, that’s nice. Okay，so that’s another difference from the conventional DFS。You do not have to go all the way to the depth。You also DFS stop either going to the maximum depth or reaching a solution。But in CSS，CSSP。

[78:00] 发言人 1: Either assign all the variable and come back, or reaches solution which is also in the maximum depth, or he stops early once he running to nothing. Okay again, these are having all input for you in the tutorial to the logo, so if you take a look, you will understand exactly what. Okay, so this is the basic for our.

[78:28] 发言人 1: So this is this forms the basic algorithm was forming a CSP. Now we move into what are the different ideas that we can make this backtracking this basic DFS much much better for our. 谢谢 people。okay，so previously。

[78:51] 发言人 1: We have seen from the green research and Acer that if you incorporating problem information to solve. Remember in the very naive case where you going on this to be a solution, if you incorporate the included on the happy distance, then the solving this will become much much better, which will be the same for CSPs. OK, so the first idea.

[79:19] 发言人 1: Now we came into our mind is does a particular order of choosing which variable to look at. Have an impact, okay? And in the original backtracking algorithm, we do not have a specific order.

[79:40] 发言人 1: We randomly pick one. We can pick Western Australia or we can pick Northern Territory wherever to start. But the question here is: is there a particular order that we can follow such that.

[79:55] 发言人 1: The efficiency of solving this problem can be quick. OK，so this is the first question. The second question is now for every variable，we have a set of available or feasible color from our feasible choices for our domain to choose from。

[80:19] 发言人 1: Previously, any DFS in backtracking we just pick any power we want, right? But describe a critical order we can pick such that we can solve this problem. Okay, another idea which actually is very common. If all these search algorithm is confused.

[80:39] 发言人 1: Okay，so later on you will see the optimization this comes up very often as well，how can you quickly terminate a search on a particular branch that you can predict or you can be sure even that it is going to be there，if it will fail, it’s not going to give you any any anything useful。Okay, so if you in the next lecture when we talk about alpha beta pooling, which is the basic idea behind alpha go, then you will see that this idea here really has been used again. In the CSP, we also show you one idea about how you can figure out these solutions early, which is be called the art consistency.

[81:19] 发言人 1: Okay，finally we will tell you how you can use problem structure, the graph structure，to make this fast。Okay，so the uh this part lecture will go into details of each one of this。Okay，so for the ordering，particularly ordering in which you visit different variable，we have a heuristic or we have an idea for the minimum remaining value。

[81:44] 发言人 1: I D okay, so what it means is that I want to choose a variable that has minimum remaining balance. Okay，so we choose the variable with the fewest the legal value left in its domain to be the first。Or to be the next variable that we want to visit.

[82:04] 发言人 1: Okay, so now in this case. Let’s say we start with. AGAIN SAYING THIS. Covering Australia is at the beginning we have several areas. None of these has been colored.

[82:28] 发言人 1: They all this value of three color that for each one of this variable. So this new problem remaining value doesn’t give you anything. Okay, let’s say we start with w equals that, and if we equal three, so we are at this point.

[82:48] 发言人 1: right here. Okay, so if you looking at, if we at this stage, the remaining variables are what? When we circle them out, After we have considered WA.

[83:06] 发言人 1: and the prince one. The only thing left for us to consider is Northern Territory, New South Wales, the Southern Australia and Tasmania. And Tasmania can immediately goes away because it’s disconnected. All these color doesn’t matter, so we will always remain as big color available.

[83:29] 发言人 1: Now，if you thinking about the northern territory，how many color there? One color，right? Because on the left and around the right，he has two neighbor states，both containing one color out，so you have to be cautious one。Same business southern Australia, he has only one color left.

[83:55] 发言人 1: Right，and for the other。For the Tasmania and for this last part right here you have three. Both for this new South Wales you have two.

[84:08] 发言人 1: Okay, so in the end I will choose either. S A. Or empty because those only with one color left.

[84:19] 发言人 1: Okay, so now you might wonder why I will always choose the variable with the minimum with the fewer legal that. 嗯。any idea.

[84:36] 发言人 1: Because if you intuitively thinking about it, choosing the variable with the minimum legal value left will give you the highest chance to fail early. Right, if you only have one value left, you just choose it. There’s nothing to choose, there’s nothing to consider.

[84:57] 发言人 1: If I as soon as choose this, if there’s any remaining variable have no legal value, then I need to be there. Right so choosing the one with minimum better left is actually give you the highest chance to fall early because that is the most constraint there. Okay，so it basically as you going down the tree，this is a day of that。

[85:28] 发言人 1: So you originally. always goes the lab. 因为我靠单他是关系维持点。

[85:38] 发言人 1: Now instead of going this way, you’re going with a park. Where you choose this such that you can return much faster. That’s where choosing this maximum value, most constraint there.

[85:58] 发言人 1: Okay，so the heuristic here we use for ordering variables is choose the variable list。The maximum, the fewest legal value left, choose the maximum legal constraint there first. Okay, now like I said, fast filling, fast filling order.

[86:20] 发言人 1: So the next question is, in my ordering, what value should I choose? So after, let’s say. Now you have you have pick a very good look at.

[86:34] 发言人 1: In here now you want to pick a color for this principle you have already picked this variable which is principle you want to pick up so for this principle I have two color options. I can either go rap, or I can go. So what is the one I should go next? What color I should expect?

[86:57] 发言人 1: Email original bfs is has go with some convention it right is on the left then i go with right if more on the left i go with. And now in this case, instead, this heuristic ask you to choose a color which give you the maximum freedom later on. That means you want to choose the value that is that is the least constraint.

[87:22] 发言人 1: Okay，so if you thinking about this. For this perspective, if you choose a red versus a blue, what will actually place out is very critical for this space. Of southern Austria, if you pick a blue here.

[87:51] 发言人 1: You basically ruled out all the options for southern Australia. Right, because the end of us know，no other things to go。So you want to pick a color。

[88:05] 发言人 1: That do not rule out as much as the other option. So the intuition behind this ordering is you want to maximize your chance. 哦，waiting to the bottom，but still satisfies all the country。Okay，so if now you choose a blue here, this branch already gets killed。

[88:29] 发言人 1: OK, so later on you will see that combining this ordering of variable and the ordering of values, basically choose a variable that hasThese remaining feasible values. Then once you’re fixing this variable, choose the color or choose the value color domain that rule outThese options for the few.Future will give you the higher chance of finding the solution faster.

[89:00] 发言人 1: And we can see that this two plus so backtracking. already solved about twenty five points. Now back training plus watering.

[89:20] 发言人 1: 给说个办法。Just by these two very simple heuristic. Right, you can imagine this just basically this heuristic function is basically returning which you already want to visit.

[89:36] 发言人 1: Okay, any question on this part? Okay，so if now let’s move on to the second main idea that we can use，which is how can we terminate a branch early。So previously。

[89:56] 发言人 1: When we are going with this, remember that tracking still at DFS right? We will either go to the end, we will either go to the end. Or we terminate at some point where we have a violation of consumer.

[90:19] 发言人 1: Right, so now the goal is. Can we go into here? And predict if we keep going we will run into violation so we stop here instead we actually see a violation.

[90:38] 发言人 1: That is basically what the future wants to achieve. Okay, so let’s take a look at how we can actually make it happen. And the idea behind this is again very simple, which is called forward chain.

[90:53] 发言人 1: So we basically wants want to run one step further of assigning a variable and see what will happen. Okay，so let’s again take this coloring of Australia as an example。Okay，so the future is basically keep track of remains for outside variable and cross off all the bad options。

[91:13] 发言人 1: Okay，so let’s start with this example。Let’s say we start with coloring the. Western Australia first, and the color this was right, and the remaining one is.

[91:28] 发言人 1: 李康文。Remaining legal options for the remaining six states. Okay, so as soon as we cover Western Australia, we know the Northern Territory only has two parts.

[91:45] 发言人 1: We agree, and we know for the Southern Australia. That also left with option of bringing in. Okay，we have seen this many times。

[92:02] 发言人 1: Now，let’s say we go to Queensland。Reason and we call it is green. So it will affect its neighboring states here, which is northern territory and southern Australia and the new world south.

[92:19] 发言人 1: Leave South West. OK. And for the northern territory, the only that was color blue.

[92:26] 发言人 1: And for South Australia. 来，包括说呢，就是。写书。

[92:37] 发言人 1: There also with color of blue right so if I keep going. For example, now I go to color this particular state, you will see that we run into a empty state here, so that is a. violation so i start.

[93:02] 发言人 1: Okay，now the question is，if you take a look at here，you may start wondering why do I have to go this. Why do I have to go one step further? Because if I’m not stopping here, I can already stop.

[93:19] 发言人 1: Can you see why I do not have to run one more step? Actually, I can stop here. Can you see why this is the case? If you take a look。Both southern Australia and northern territory only left with color blue.

[93:45] 发言人 1: And if you’re seeing here, they’re actually neighboring two states. So as soon as they are running here. I could actually start.

[93:55] 发言人 1: Because I know, doesn’t matter what color I assign. For example, the next. I can assign this, I can assign this, I can assign that.

[94:03] 发言人 1: But doesn’t matter what I assign the end is going to break. This branch doesn’t mean doesn’t make me anywhere as soon as I place the green to Brisbane and the raft to Western Australia. And this particular combination has no visible.

[94:19] 发言人 1: Doesn’t matter what the rest of this. correct you put. Right, so the idea is can be stop here.

[94:28] 发言人 1: If so, how? Okay，and idea here is called a constraint propagation。Okay，so when we actually assign a color，we want forward checking。If this is going to bring any other constraint here. In my problem, okay? So this is why just show you as soon as you going into this particular space, if you do a.

[95:00] 发言人 1: Forward propagation checking on the constraint，you will know that these two cannot be simultaneously satisfied，so without you coloring Tasmania, without you coloring the uh New South Wales，without you coloring any other places，you know these two cannot both share blue，so you should try。Okay，and the way we do this is called constraint propagation，and let me give you the exact algorithm on how to make this happen。Okay，so before we looking at the actual algorithm，let us first define a concept is called the consistency of an art，or in short，art consistency。

[95:45] 发言人 1: Okay，so why this is called an artist? If you imagine previously we talk about，we can represent this CSP as a graph，and this art basically tells you there’s a constraint between any two variables。So this is our consistency basically check if these two constraint pairs have any feasible value right?

[96:10] 发言人 1: Okay，this is the conceptually，but if you put this into an algorithm, what it means is that if there’s a art。Coming from x variable pointing to y variable where we call this x as my head. ok，嗯，excuse me，i may in this case。If I call my access my pet.

[96:44] 发言人 1: And the why is my what I want to do is I want to check every option. in my pocket. If there is some value in my head, why such that? I know，我也有那个课程。

[97:07] 发言人 1: Okay，so now let’s take an example. Let’s say my tail is my W A, and my hat is my own territory. So you have a heart goes like this. Okay，so what it actually says is for every x in my hat，for every x in my tale，so now I have already assign WA to red，so this x only。来，color is right。

[97:40] 发言人 1: now i want to check in my tab，which is。I mean in my head this is my l in my head and t do i have a color such that i do not validate this fact so originally. I have red, green and blue.

[98:06] 发言人 1: Since my title company already assign color red that means. If I ignore this red, I have at least two color in my head that do not violate this. It tells color that.

[98:22] 发言人 1: Okay，so this part is consistent。Now, if I go the other way around, instead of coming from WA to Northern Territory, if I now go from Northern

Territory to WA, that means my Northern Territory become2. And my W A become happy again going with this definition I want to check every color.

[98:54] 发言人 1: In my empty whether I can have a color that in my head that’s not violated. So I start with color. Maybe i start color.

[99:11] 发言人 1: is there any color in my WA but not violate my color red don’t do. The answer is no, because WA already being assigned color with red. So the only option for WA is red, which is in conflict with this color red.

[99:30] 发言人 1: If I choose red for the territory, so if I wrong our consistency, I will immediately remove this red point because this red color is no longer consistent. Now I will move on to the second color of green. It’s very color from my hat WA, which not violating with my choice of color green.

[99:52] 发言人 1: The answer is yes, so I keep it. Now I go to. Check if there’s a option in my W eight, the not violated color is blue here. Yes, the red is not violated is blue, so my blue color here also stay.

[100:11] 发言人 1: Okay，so so on and so forth，you can do this with W A to S A and S A to W A，you will see that the red color of my essay also goes away。And if you do any other pair empty to Q Q to empty S A to this this to that you will see that no other color will be from this constraint operation. Okay，so in the end，you will resolve in the after you are running with all these pairs。

[100:45] 发言人 1: You can see that if for the whole graph there exists some colors that do not violate the rest of the uh constraints. Ok, so if you do this, you will resulting if you can. If this constraint is can be satisfied for any such art here, from x to y, x can be anything, for all variable y also can be anything from my seven variable, if you can do this, you will say the whole graph is art consistent.

[101:16] 发言人 1: Okay，so if you do this propagation，let’s say for me Victoria to New South Wales，and do all this there，in the end you can have different。results after you running this are consistency but if you what i want to show you is previously. 嗯，你好。

[101:41] 发言人 1: In the previous case, without this filtering, you still need to further assign a color before you can stop. But and here, if you run this arc consistency, you will find that if your. going for MT to SA.

[102:04] 发言人 1: If you run this consistency, you will find that you cannot satisfy this. Right so if you medium if you cannot satisfy any such part in your constraint graph you can terminate her. Okay，so this is basically the whole idea behind this art consistency。

[102:27] 发言人 1: I already know，but again，if you want to implement this is the student。Okay，so we have also input in this for you in the Jupiter notebook。So all we want you to try to do is check to see if you can actually reproduce all this，or you can run the step by step to see if this is producing the result illustrated。

[102:52] 发言人 1: In the classroom. Okay，so something I want you to actually. Notice is that the R consistency is really expensive algorithm. He actually runs with every single pairs.

[103:10] 发言人 1: In the ground, so it’s actually in the order of square in terms of number of notes, and in the order of cubic in terms of number of options you have for the rest. So in the end this can be a really really expensive algorithm. So one of the immediate question in your mind will comes from this.

[103:32] 发言人 1: So now let’s say you are at the states. You at least have two options. One is run our consistency to see if this can be terminated or not.

[103:46] 发言人 1: Before running our consistency, you don’t know. The second is you do not run our consistency and just run it up. Keep running it and terminating after you see a violation. Okay, so now we come. Complexity resulting that you’re trying to compare this the complexity of keep running this which is the complexity of that first search versus the complexity of our systems. Okay， so if you take a look at the complexity of the art consistency，if you have a large number of n and a large number of d，our consistency may be even expensive，more expensive than keep running a dfs，meaning that you cannot love the art consistency，you can’t even solving this。

[104:44] 发言人 1: Okay，so all this over tracking is good。That can help you to solve this problem in minimum states and visit but in physical time and physical complexity you may not be the optimal. Okay，and there’s one more very important。

[105:06] 发言人 1: Problem that comes with our consistency is a whole graph satisfies our consistency. That’s not means that we will guarantee a feasible solution. okay, so let me give you this following example。

[105:27] 发言人 1: If you see this graph, it is actually a art assistant graph. If you see this angle, for any color you pick on the tail, there is a color that do not like. If you pick this.

[105:45] 发言人 1: 啊，对对的。If you do this here, this here, this here, this here, this here, you all the same. So this graph is a art consistent graph.

[105:59] 发言人 1: Right, so it is okay by the artist is reality, but if you imagine now. as soon as you. Pick the choice.

[106:17] 发言人 1: For example, now you pick red and remove this. 嗯，你配个软件。Then the only option left for this one is me.

[106:31] 发言人 1: Now you will run into a violation. Which is true for any other combination you choose. 你是不缺心的。应该其实。

[106:45] 发言人 1: Now you will eliminate in this blue, and you have right here. Now you have run up options. So what I want to show you through this example is you can have a our consistent graph.

[107:00] 发言人 1: It does not guarantee in the end you have a feasible solution. Okay, so you go through this and square be two bit complication of running all these art consistency in the end，is still do not guarantee you a feasible solution。Okay，this is something that you should be careful when running this algorithm.

[107:25] 发言人 1: Okay，so this is the second idea that we can try. The first idea is fix in which order I visit variables and in which order I visit the available options from the domain. That improves at least two order of magnitude in terms of effectiveness and in terms of size of the problem I guess so.

[107:46] 发言人 1: Then we going into this futureWhich can actually help you and detect long with long good path early on. OK, but this effi- efficiency is really low because. It is actually a swear to the problem size, to the number of variable in a cube, to the number of options.

[108:08] 发言人 1: That’s why later on, if you see the AlphaGo paper, they actually do not want that our consistency. What they want is from a user level. Okay，so later on we will talk touch about this when we talk about elevator，okay? So now the last part I want to talk about is from the card structure，this is really coming from the search space based graph，right?

[108:35] 发言人 1: So given this structure，what are the information that we can use to solve this? Okay，so the very very first idea that we have，which is actually reflecting the real world problems in reality，we most likely do not have a fully connected graph，that means the whole graph is connected to the rest of the complete。There is no independent problems.

[109:02] 发言人 1: Okay, you most likely in real world many problems can be decomposed into subproblems. In this case, you can see the Tasmania is actually an independent problem from the rest. So why is this useful? So previously, we know the whole problem has very believable set.

[109:28] 发言人 1: But in this case, we actually correspond to two problem. One problem has variable six, the other problem has variable one. You can actually independently swaps.

[109:45] 发言人 1: Okay，the solution of this problem does not affect the solution of this problem。Likewise，the solution to this problem will not affect the problem to the right。Okay，so using this subproblem。

[110:00] 发言人 1: Independence structure can help you decompose the solving of this problem into two independent problems. You can solve that in parallel, or you can solve it sequentially. It does not change the solution of this problem.

[110:15] 发言人 1: Okay, so now if we do the calculation, if the original problem has been and variables, if it can be divided into. Some problems only containing C there the concrete cases let’s say now if the problem has eight variables. Each variable is a binary variable, can only choose two values, in the zero one, and in the end you can factor this into some problems where you only have twenty variables, then the original.

[110:53] 发言人 1: Total number of total number of configuration that you need to consider is two raised to the power of eight. OK，which take more than billions of years to solve with most advanced computer。But now if you can somehow.

[111:13] 发言人 1: We compose this problem into sub problems where there’s at most body variables. Let’s say in this case that we resolve in four sub problems,

then the resulting number of total number of states can be solved in less than a second. With the same amount of compete, so given the same computer, the original problem took four billion years to solve.

[111:39] 发言人 1: The second become most probably to the same. Okay, so this is a huge difference in terms of the size of the product, okay, so if it naturally has independent structure。That we can leverage, okay? And the second problem structure is originally the problem has been put as a graph.

[112:07] 发言人 1: However, if the graph structure. Of the problem can be represented as the two. Okay, remember when we talking about these three search algorithm on the left, we always show you a graph.

[112:26] 发言人 1: On the right, we always show you a tree. Okay, so if this graph can be factored into a tree. Then there's some heuristics that we can use that can help you previously show to solve this.

[112:40] 发言人 1: The original graph is n squared in q, but if you can factor this graph into a tree, then you can reduce the computation time to n times in square. Okay, so two order of magnitude more efficient than the original population。Okay, this is another very important problem structure that you can leverage, or if your graph can be factor in a tree, then we can do this。

[113:11] 发言人 1: okay then the second. Structure, our structure that we can leverage is called approximation。Okay, so in this case, if you're thinking about the color in Australia case, the complication happens with the southern Australia, which every no sort of connect through this.

[113:37] 发言人 1: 因为这个 state neighbor many other notes。And this really makes this graph cannot be converted into a single tree. And this is where it makes all these constraints propagation etc.

[113:53] 发言人 1: Okay, so what we can do is we can do approximation, we can try read this。As if the graph be be South Australia know is not there, then how this graph can be solved? Okay, so this realization, really this approximation of the grass structure can again help you reduce the complexity。

[114:20] 发言人 1: Okay, so because because of time issues in the past, I'm not going to elaborate on this, but in the tutorial we have one implementation for you, which you can try that out yourself。Okay, so this concludes everything in terms of the material I want to cover. So in the first part, we talk about how we can solve the CSP in general.

[114:44] 发言人 1: Where the basic idea is to use depth first search, because we know the breath first search doesn't work at all. OK, then we talk about three specific techniques on how to make it faster. First is order, because.

[115:04] 发言人 1: 我们 just pick some, no other left, who knows whether that is true, so fix gave a heuristic to reach order works well。Okay, second is how can we terminate a no promising branch earlier? That is the filtering, the art consistency algorithm。

[115:24] 发言人 1: The third is if we looking at the space space graph of the resulting problem，the CSP is there any problem structure? Now we can directly laugh. Okay, so these three together help us to solve the problem much much better。

[115:42] 发言人 1: If you try this on any problem，that can be solved with head down the face easy。It’s all these three together。Okay，so in the tutorial we gave the example for you to try it on yourself。

[115:54] 发言人 1: Okay，so what we will do next is we can take another ten minutes break。When we come back, we have prepared some real time exercise for you to try. So we will together actually some exercise about the search problem as well as the constraint satisfaction problem that we together try in class.

[116:14] 发言人 1: Okay, so let’s take a quick10 minutes break. We will resume around11:40 where we will try some problems, some handwriting problems. Okay。